LI, Yan

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EDUCATION

The University of Wisconsin, Madison 09/2025 - 06/2027

Master of Arts in Foundation of Advanced Studies | Department of Mathematics

GPA: **3.917/4.0** | **Coursework**: Real Analysis, Stochastic Calculus, Partial Differential Equations

The Chinese University of Hong Kong, Shenzhen 09/2020 - 06/2025

Bachelor of Science, Major in **Mathematics (Applied Mathematics)** 09/2023 - 06/2025

Major in **Financial Engineering** 09/2020 - 06/2023

Awards: Dean's List (2021-22, 2022-23, 2023-2024, 2024-2025), Master's List (2021-22)

RESEARCH EXPERIENCE

Research Assistant| Computational Imaging Group (CIG), UWisc 09/2026 - Present

Topic: Image Denoising with Dispatch Diffusion Models | **Advisor**: Prof. Ulugbek S. Kamilov

- Conducting research on image denoising using dispatch diffusion models, with a focus on improving robustness and reconstruction quality for computational imaging applications.
- Exploring diffusion-based inverse problem-solving frameworks for imaging data, including model implementation, experiment design, and quantitative evaluation of denoising performance.

Visiting Student Researcher | Shenzhen Research Institute of Big Data 06/2023 - 12/2023

Topic: AI-Driven Innovations in Medication Adherence Monitoring and Scleral Jaundice Detection

Advisor: Dr. Li, Haofeng

Stage 1: Medication Adherence Monitoring Project

- Implemented YOLOv8 for pill detection to support technology-driven monitoring of medication adherence among chronic disease patients.
- Configured environments, annotated video data, and reproduced MagNet and HRNet segmentation models, improving IoU through failed-case analysis.

Stage 2: Scleral Jaundice Detection and Facial Privacy Protection

- Expanded a Class I scleral jaundice dataset and trained YOLOv8 to establish a stronger baseline for liver-function monitoring applications.
- Developed hand and pose landmark models to support facial privacy protection for scleral jaundice detection scenarios.

Leader | DDA4210: Advanced Machine Learning 02/2023-05/2023

Topic: Fake Face Detection: Enhancing Accuracy with GAN model and Iris Color Analysis

Advisor: Prof. Li, Tongxin

- Led a course project on AI-generated face detection by developing a GAN inversion-based model and iris color comparison method, improving accuracy by 10% while evaluating fairness across race and gender groups.

Member | ERG3020: Web Analytics and Intelligence 02/2023-05/2023

Topic: MBTI Classification of Novel Characters Based on PageRank and LSTM

Advisor: Prof. Tang, Xiaoying

- Built a PageRank- and LSTM-based framework to predict MBTI types of novel characters, using centrality and dialogue length to support personality analysis and recommendation applications.

LEADERSHIP EXPERIENCE

President, Han Culture Club | **President**, Badminton Club | **Founder**, Siyue Culture Club

Head of Action Department, CUHK Animal Protection Association | Rescued 20+ stray cats in collaboration with pet hospitals; raised 10,000+(RMB) to support animal welfare.

Student Assistant, CUHK Career Development Center | Coordinated 10+ events, including career fairs and workshops; managed daily website updates, increasing traffic to 10,000+ views/day.

Leader, Wand Startup Team | Explored ChatGPT's NLP capabilities and developed GPT-4 API-based applications for personalized academic tutoring and psychological counseling, while drafting the business plan and shaping the team's business strategy.